

Section M

Differential equations

$$1. y = \int \frac{x}{2} dx = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)x^2 + c$$

$$y\left(\frac{1}{2}\right) = -1 \Rightarrow \frac{1}{4}\left(\frac{1}{2}\right)^2 + c = -1$$

$$\Rightarrow c = -1 - \frac{1}{16} = -\frac{17}{16}$$

$$\therefore y = \frac{1}{4}x^2 - \frac{17}{16}$$

$$2. y = \int -\frac{3}{2}x^2 = \left(-\frac{3}{2}\right)\left(\frac{1}{3}\right)x^3 + c$$

$$y(-1) = -\frac{1}{2} \Rightarrow \frac{1}{2} + c = -\frac{1}{2}$$

$$\Rightarrow c = -1$$

$$\therefore y = -\frac{1}{2}x^3 - 1$$

$$3. y = \int \sin t dt = -\cos t + c$$

$$y\left(\frac{\pi}{2}\right) = -\cos \frac{\pi}{2} + c = 3$$

$$\Rightarrow c = 3$$

$$\therefore y = -\cos t + 3$$

$$4. y = \int e^x dx = e^x + c$$

$$y(0) = e^0 + c = 4$$

$$\Rightarrow c = 3$$

$$\therefore y = e^x + 3$$

$$5. y = \int \frac{1}{x} dx = \ln|x| + c$$

$$y(1) = \ln 1 + c = 2$$

$$\Rightarrow c = 2$$

$$\therefore y = \ln|x| + 2$$

$$6. \frac{dy}{dx} = \int 0 dx = c$$

$$y'(2) = 1 \Rightarrow c = 1$$

$$\therefore \frac{dy}{dx} = 1$$

$$y = \int 1 dx = x + c$$

$$y(2) = 2 + c = 2 \Rightarrow c = 0$$

$$\therefore y = x$$

$$7. \frac{dy}{dx} = \int \cos x dx = \sin x + c$$

$$y'(0) = \sin 0 + c = 1 \Rightarrow c = 1$$

$$\therefore \frac{dy}{dx} = \sin x + 1$$

$$y = \int (\sin x + 1) dx$$

$$= -\cos x + x + c$$

$$y(0) = -\cos 0 + 0 + c = 2$$

$$\Rightarrow c = 3$$

$$\therefore y = -\cos x + x + 3$$

$$8. \frac{dy}{dt} = \int e^t dt = e^t + c$$

$$y'(1) = e + c = e \Rightarrow c = 0$$

$$\therefore y' = e^t$$

$$y = \int e^t dt = e^t + c$$

$$y(1) = e + c = -4e$$

$$\Rightarrow c = -5e$$

$$\therefore y = e^t - 5e$$